Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	*(a)*	(i)		all four carbonates (1) acids react with metal carbonates to form carbon dioxide gas (1)	2			2		2
		(ii)		(hydrochloric acid would form) <u>insoluble</u> compound with Pb^{2+} (1) should use nitric acid / ethanoic acid (1)			2	2		2
		(iii)		award (2) for **all** four correct award (1) for **any** two correct $Mg^{2+}(aq)$ white (precipitate) $Fe^{2+}(aq)$ (dark) green (precipitate) $Cr^{3+}(aq)$ (grey) green (precipitate) $Pb^{2+}(aq)$ white (precipitate)	2			2		2
		(iv)		add excess sodium hydroxide (1) award (1) for either of following • magnesium hydroxide white precipitate remains but lead hydroxide precipitate dissolves (giving a colourless solution) • iron(II) hydroxide green precipitate remains but chromium hydroxide precipitate dissolves (giving a dark green solution) lead and chromium are amphoteric (iron and magnesium are not) (1)			3	3		3

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
	(b)	(i)		Mg^{2+} $\frac{91}{24.3}$ = 3.7449 Ca^{2+} $\frac{50}{40.1}$ = 1.2469 (1) ratio is 3Mg: 1Ca (1) formula must contain 4 × CO_3 so it is $Mg_3Ca(CO_3)_4$ (1) accept alternative correct representations e.g. $CaCO_3.3MgCO_3$ $(MgCO_3)3.CaCO_3$		1	2	3	2	
		(ii)		concentration = 1.25 × 10^{-6} mol dm^{-3} (from calcium or magnesium concentrations) (1) moles of solid used = $\frac{220 \times 10^{-6}}{353}$ = 0.6232 × 10^{-6} mol (1) volume = 0.50 dm^3 (1) allow any value in the range 0.498-0.501 dm^3		1 1	1	3	2	
				Question 12 total	**4**	**3**	**8**	**15**	**4**	**9**